# Fabric Steamer



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

**Covers these lab items:** Laboratory Fabric Streamer

## What it is used for

A fabric steamer applies controlled steam to textile samples or materials to assess or improve properties such as shrinkage, colour fastness, crease removal, or dimensional stability, depending on the equipment and test method. Correct temperature, steam exposure, and sample handling help achieve repeatable results while reducing burn and fabric-damage risks.

## Main parts

- Steam chamber or steaming enclosure
- Steam generator or approved steam supply
- Fabric sample holder, rack, or tray
- Temperature and pressure controls or indicators
- Condensate drain and steam outlet
- Timer, lid, seals, and protective housing

## Before you start

1. Confirm the intended process or test method, fabric type, sample dimensions, and required steam exposure.
2. Read the equipment manual and verify that the steamer is suitable for the fabric and the planned procedure.
3. Inspect the chamber, seals, sample holders, drain, steam hoses, and controls for damage or leaks.
4. Check the water supply or approved steam connection and confirm that the drain is clear.
5. Verify that temperature and pressure indicators, interlocks, and over-temperature protection are functioning as specified.
6. Wear suitable heat-resistant gloves, eye protection, and other required PPE; prepare a heat-resistant surface for handling samples.

## How to use

1. Place the steamer on a stable surface with adequate ventilation and clear access to its controls.
2. Fill the reservoir with the specified water or connect the approved steam supply according to the manual.
3. Set the required temperature, steam level, and exposure time according to the approved procedure.
4. Switch on the unit and allow it to reach the specified operating conditions before loading the sample, where required.

5. Prepare and label the fabric sample, then position it on the designated holder without blocking steam outlets.
6. Close the chamber or lid securely and confirm that seals and safety interlocks are engaged.
7. Start the steaming cycle and monitor the indicated temperature, pressure where applicable, and remaining time.
8. Avoid opening the chamber during the cycle unless the procedure specifically requires it and the equipment can be opened safely.
9. Stop the process if a steam leak, abnormal pressure, control fault, or unexpected temperature rise occurs.
10. At the end of the cycle, switch off steam generation and follow the manufacturer's depressurisation and cooling instructions.
11. Open the chamber only when the indicated conditions and safety interlocks permit; stand clear of escaping residual steam.
12. Remove the sample using heat-resistant gloves or approved tools and place it on a safe surface to cool or condition as required.
13. Inspect the sample and record the test conditions, exposure time, and observed changes according to the procedure.
14. Drain and clean the unit after cooling as instructed, then switch off and isolate power if required.

## ⚠ Safety

- Only trained personnel should operate the fabric steamer and interpret formal test results.
- Steam and hot condensate can cause severe burns; keep hands and face away from steam outlets and the chamber opening.
- Never open a pressurised chamber or bypass lid interlocks and pressure-relief devices.
- Do not exceed the equipment's rated temperature, pressure, or exposure limits.
- Use only the specified water and approved steam supply; keep electrical connections dry.
- Allow the unit to cool and depressurise before cleaning, draining, or maintenance.
- Stop use and report any damaged seals, leaking hoses, faulty controls, or abnormal pressure.

## Care & storage

- Drain condensate and clean the chamber, sample holders, and accessible surfaces after cooling.
- Inspect seals, hoses, fittings, and the drain system regularly for wear, scale, or leaks.
- Remove scale using only products and methods approved by the manufacturer.
- Check temperature and pressure indicators and arrange calibration where required.
- Store the steamer clean and dry, and arrange servicing if steam delivery or temperature control becomes unreliable.

## Common mistakes to avoid

- Using an incorrect steam temperature or exposure time for the fabric or test method.
- Opening the chamber before steam pressure and temperature reach a safe condition.
- Placing samples where they block steam circulation or contact hot surfaces unexpectedly.
- Ignoring scale build-up, blocked drainage, or leaking seals.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*



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